01 · Introduction to Angular

Beginner Angular learning notes

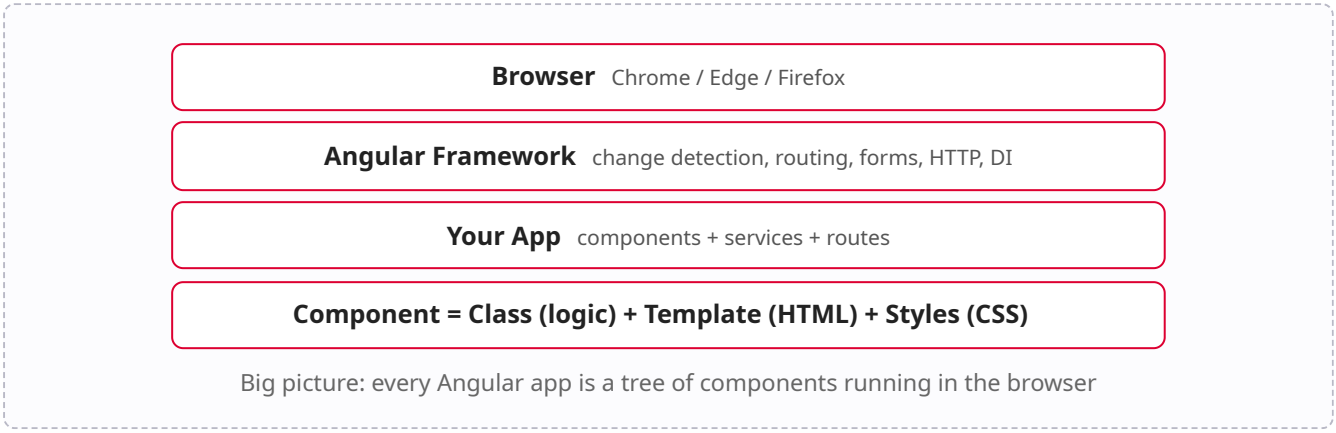
a. Definition

Angular is a free, open-source framework from Google for building web applications using **TypeScript**, HTML and CSS. Instead of writing one huge page, you build small reusable pieces called **components** and snap them together like LEGO blocks.

Angular apps are usually **Single Page Applications (SPA)**: the browser loads the page once, and after that Angular only swaps the parts of the screen that change. No full page reload, so the app feels fast like a mobile app.

Angular is a complete platform: it ships with routing, forms, HTTP, testing tools and a command-line tool (the Angular CLI), so you do not have to pick and glue together many separate libraries.

b. Diagram



Big picture: every Angular app is a tree of components running in the browser

c. Syntax

```
@Component({
  selector: 'app-name',        // the HTML tag you will use
  template: `<h1>...</h1>`,    // what to show (or templateUrl: './x.html')
  styles: `h1 { color: red }`, // how it looks (or styleUrl)
})
export class NameComponent {
  // data and methods for the template
}
```

d. Example Program

`Programs/01_Introduction_to_Angular/hello-world.component.ts`

```
import { Component } from '@angular/core';

// A component is a TypeScript class + an HTML template + CSS styles.
@Component({
  selector: 'app-root', // the tag <app-root> in index.html will show this component
  template: `
    <h1>{{ title }}</h1>
```

```
    <p>{{ message }}</p>
  `,
  styles: `
    h1 { color: #dd0031; }
  `,
})
export class HelloWorldComponent {
  // Properties of the class can be shown in the template with {{ }}
  title = 'Hello, Angular!';
  message = 'My first Angular component is running.';
}
```

e. Output

Rendered in the browser (text on screen):

```
Hello, Angular!
My first Angular component is running.
```

f. Real-World Example (English + தமிழ்)

English

Think of a railway station display board: only the train timings change on the screen, the whole building is not rebuilt. An Angular single page app refreshes only the part of the page that changed.

தமிழ்

சென்னை மெட்ரோ ரயில் நிலையத்தில் உள்ள அறிவிப்புத் திரையை நினைத்துப் பாருங்கள். அடுத்த ரயில் எப்போது வரும் என்ற நேரம் மட்டும் மாறுகிறது; முழு நிலையக் கட்டிடமும் மீண்டும் கட்டப்படுவதில்லை. அதே போலத்தான் Angular-இல் (SPA) பக்கம் ஒரு முறை மட்டுமே ஏற்றப்படும், அதன் பிறகு மாற வேண்டிய பகுதி மட்டும் புதுப்பிக்கப்படும்.

அந்தத் திரையில் உள்ள ஒவ்வொரு பகுதியும் - ரயில் பெயர், நேரம், நடைமேடை எண் - ஒவ்வொரு தனித் **component** போல. பல component-களை ஒன்றாகச் சேர்த்துதான் முழு Angular செயலியை உருவாக்குகிறோம்.

g. Pros

- Complete framework (router, forms, HTTP)
- Strong TypeScript tooling
- Backed by Google, long-term support
- Good for large teams

h. Cons

- Steeper learning curve than React or Vue
- More boilerplate to learn
- Larger initial bundle for tiny apps

i. Advanced / New Features

- Standalone components are the default (v19+)
- Signals for reactivity
- New `@if/@for` control flow
- Zoneless change detection (default in new v21 apps)
- Vite/esbuild based builds

j. Common Mistakes

- Confusing **AngularJS** (old, 2010, JavaScript) with **Angular** (2016+, TypeScript). They are different frameworks.
- Thinking you must learn everything at once. Start with components and templates; the rest comes step by step.
- Skipping TypeScript basics, then getting stuck on error messages.
- Trying to edit the DOM directly (`document.getElementById`) instead of using template bindings.

k. Interview Questions

Q1. What is Angular?
A TypeScript-based, open-source framework by Google for building single page web applications using components.

Q2. What is the difference between AngularJS and Angular?
AngularJS (1.x) is JavaScript with controllers and scopes. Angular (2+) is a TypeScript rewrite built on components, with better performance and tooling.

Q3. What is a Single Page Application?
An app that loads one HTML page and updates parts of it dynamically using JavaScript instead of reloading full pages.

Q4. Name the main building blocks of Angular.
Components, templates, directives, pipes, services with dependency injection, modules/standalone components and the router.

l. Practice Exercises

1. Write down 3 websites you use daily that behave like a SPA (no full reload when you click around).
2. In `hello-world.component.ts`, change the title and message, then add a second `<p>` using a new property.

3. Draw (on paper) the component tree for a shopping website: header, product list, product card, cart.